

Probability and Statistics

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ES-III Spring 2025 Assignment # 5

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SECTION : CE (T) =

Q1. The #s below represent the number of sick days claimed :

15, 7, 8, 95, 19, 12, 8, 22, 14

$$\Rightarrow \text{Mean} = \frac{15 + 7 + 8 + 95 + 19 + 12 + 8 + 22 + 14}{9}$$

$$\text{Mean} = 22.22$$

$\Rightarrow$ Median : in order to find the median, the data needs to be sorted in descending order (smallest $\rightarrow$ largest) ;

7, 8, 8, 12, 14, 15, 19, 22, 95

Since there are 9 values (an odd number), the median is the middle value (5th one) :

14

$\Rightarrow$ Mode : the value that appears the MOST FREQUENTLY :
8

Looking at the data, 95, is much larger than the rest. It is an outlier, $\Rightarrow$ Outliers heavily affect the mean.

$\therefore$ The median is the best measure of center because it is not affected by the outlier (95) and better reflects the middle of the dataset.

Q2. Tar contents of 8 brands:

7.3, 8.6, 10.4, 16.1, 12.2, 15.1, 14.5, 9.3

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a) Mean = $\frac{7.3 + 8.6 + 10.4 + 16.1 + 12.2 + 15.1 + 14.5 + 9.3}{8}$

8

Mean = 11.6875 milligrams

b) 8 brands of cigarettes are selected at random;

this implies that the 8 brands are a sample taken from a larger population of cigarette brands.

$\therefore$ sample variance: $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$; $\bar{x} = \text{sample mean} = 11.6875$

$\Rightarrow s^2 = \frac{(7.3-11.7)^2 + (8.6-11.7)^2 + (10.4-11.7)^2 + (16.1-11.7)^2 + (12.2-11.7)^2 + (15.1-11.7)^2 + (14.5-11.7)^2 + (9.3-11.7)^2}{8-1}$

8-1

$s^2 = \frac{75.42}{7} = 10.77$ milligrams sq

Q3. Score differences; $n = 13$

a) Mean = $\frac{11 + 49 + 32 + 3 + 6 + 38 + 38 + 30 + 8 + 40 + 31 + 5 + 36}{13}$

Mean = 25.15

b) Median: 3, 5, 6, 8, 11, 30, 31, 32, 36, 38, 38, 40, 49

$\Rightarrow$ Since there are 13 values (odd number), the median is the 7th value.

31

Q4. $n = 16$

$\mu = 50$ } population

$\sigma = 5$ } statistics

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} = \frac{(5)^2}{16} = 1.5625$

$\sigma_{\bar{x}} = \sqrt{1.5625} = 1.25$

Since all the possible samples of size 16 are drawn from a normal distribution;

$$\mu_{\bar{x}} = \mu$$

$$\Rightarrow \mu_{\bar{x}} = 50$$

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$$\therefore \text{Lower bound} = \mu_{\bar{x}} - 1.9 \sigma_{\bar{x}}$$

$$= 50 - 1.9 (1.25)$$

$$= 50 - 2.375$$

$$= 47.625$$

$$\text{Upper bound} = 50 - 0.4 (1.25)$$

$$= 50 - 0.5$$

$$= 49.5$$

$$\Rightarrow P(47.625 < \bar{x} < 49.5)$$

$$\Rightarrow z_1 = \frac{47.625 - 50}{1.25} = -1.9$$

$$\Rightarrow z_2 = \frac{49.5 - 50}{1.25} = -0.4$$

$$\Rightarrow P(-1.9 < Z < -0.4)$$

$$\Rightarrow P(Z < -0.4) = 0.3446$$

$$P(Z < -1.9) = 0.0287$$

$$\therefore P(-1.9 < Z < -0.4) = 0.3446 - 0.0287 = 0.3159$$

This means there is a 31.59 % chance that the sample mean $\bar{x}$ will fall in that interval.

5. Random sample of size 54 ;

$$n = 54$$

$$\Rightarrow n > 30$$

$\therefore$ The population and the sampling distribution of the sample mean are all normally distributed ;

$$\Rightarrow P(4.1 < \bar{x} < 4.4)$$

For the mean, μ : $\mu_{\bar{x}} = \mu$

$\Rightarrow$ Converting to z-scores :

$$\Rightarrow \mu = \frac{2+4+6}{3} = 4 = \mu_{\bar{x}}$$

$$\Rightarrow z_1 = \frac{4.1 - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

For the standard deviation of the

$$\Rightarrow z_2 = \frac{4.4 - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

$$\text{population : } \Rightarrow \frac{\sum (x_i - \mu)^2}{n} = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3}$$

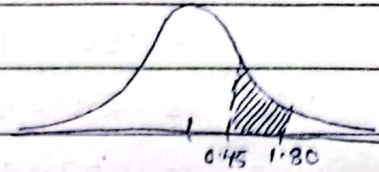
$$\Rightarrow \frac{8}{3} = 2.667 = \sqrt{2.667} = 1.63 = \sigma$$

$$\therefore \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.635}{\sqrt{54}} = 0.222$$

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$$\therefore z_1 = \frac{4.1 - 4}{0.222} = 0.45$$

$$z_2 = \frac{4.4 - 4}{0.222} = 1.80$$



$$\Rightarrow P(0.45 < Z < 1.80) = 0.9641 - 0.6736 = 0.2905$$

$\Rightarrow$ 29.05 % chance.

Q6. Population of students.

$$\Rightarrow \begin{cases} \mu = 174.5 & \text{population} \\ \sigma = 6.9 & \text{stats.} \end{cases}$$

$$\Rightarrow n = 25$$

$$\Rightarrow n < 30$$

Since $n < 30$, and the population is normally distributed, the sampling distribution of the sample mean is also normally distributed.

$$\therefore \mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

a) Mean of the sampling distribution, $\bar{x}$, $\mu_{\bar{x}} = \mu$

$$\mu_{\bar{x}} = 174.5 \text{ cm}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6.9}{\sqrt{25}} = 1.38 \text{ cm}$$

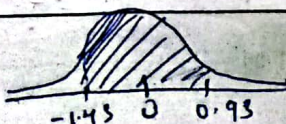
$$b) P(172.5 \leq \bar{x} \leq 175.8)$$

$\Rightarrow$ Converting to z-scores;

$$\Rightarrow z_1 = \frac{172.5 - 174.5}{1.4} = -1.42857 = -1.43$$

$$\Rightarrow z_2 = \frac{175.8 - 174.5}{1.4} = 0.92857 = 0.93$$

$$\Rightarrow P(-1.43 \leq Z \leq 0.93) = 0.8238 - 0.0764 = 0.7474$$



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$$\therefore 0.7474 \times 200$$

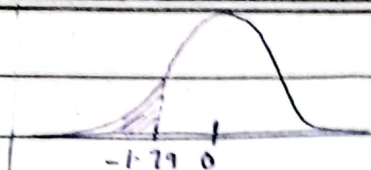
$$= 149.48 \approx 149 \text{ samples.}$$

$$c) P(\bar{y} < 172)$$

$$\Rightarrow z_1 = \frac{172 - 174.5}{1.4} = -1.7857 \approx -1.79$$

0.44)

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$$\Rightarrow P(Z < -1.79)$$

$$\Rightarrow 0.0367$$

$$\therefore 0.0367 \times 200 = 7.34 \% \text{ chance}$$

$$\mu = 40$$

$$\sigma = 2$$

$$n = 36$$

The combined resistance of 36 resistors is the sum of the sample:

$$\Rightarrow \mu_{\text{sum}} = n \cdot \mu = 36(40) = 1440$$

$$\Rightarrow \sigma_{\text{sum}} = \sqrt{n} \cdot \sigma = \sqrt{36}(2) = 6(2) = 12$$

$$\Rightarrow P(\text{sum} \geq 1458) \Rightarrow Z = \frac{1458 - 1440}{12} = \frac{18}{12} = 1.5$$

$$\Rightarrow P(Z > 1.5) = 1 - P(Z \leq 1.5) = 1 - 0.9332 = 0.0668$$

$$\Rightarrow 6.68 \% \text{ chance}$$

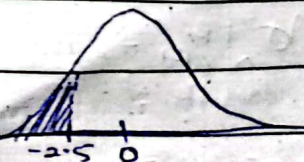
$$\mu = 3.2 \text{ mins}$$

$$\sigma = 1.6 \text{ mins}$$

$$n = 64$$

$$P(\bar{x} \leq 2.7) \text{ } \} \text{ at most 2.7 minutes;}$$

$$\Rightarrow Z = \frac{2.7 - 3.2}{1.6/\sqrt{64}} = -2.5$$

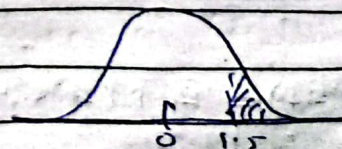


$$\therefore P(Z < -2.5) = 0.0062$$

$$\Rightarrow 0.62 \% \text{ chance}$$

$$P(\bar{x} > 3.5)$$

$$\Rightarrow Z = \frac{3.5 - 3.2}{1.6/\sqrt{64}} = 1.5$$



$$\Rightarrow P(Z > 1.5) =$$

$$1 - P(Z \leq 1.5) = 1 - 0.9332$$

$$= 0.0668 \times 100 = 6.68 \% \text{ chance}$$

$$(C) \quad P(3.2 < \bar{X} < 3.4)$$

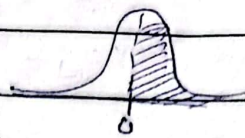
$$\therefore \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \Rightarrow z_1 = \frac{3.2 - 3.2}{1.6/\sqrt{64}} = 0$$

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$$\Rightarrow z_2 = \frac{3.4 - 3.2}{1.6/\sqrt{64}} = 1$$

$$\therefore P(0 < Z < 1)$$

$$\Rightarrow 0.5 \times 100 = 50\% \text{ chance.}$$



Q9. $\mu = 540$

$$\sigma = 50$$

We define two sample sizes;

$$n_1 = 32$$

$$n_2 = 50$$

a) $P(\text{score } (\bar{X}) \text{ difference} > 20)$

Since both of the sample sizes are > 30 , $\therefore$ the sampling distribution of the sample mean follows a normal distribution.

$$\Rightarrow \text{for } n_1 = 32;$$

$\Rightarrow$ Let d be the difference between the mean scores of both the sample sizes;

$\Rightarrow$ Because both the samples follow normal distributions, the difference will also follow a normal distribution.

$$\Rightarrow \text{The mean difference will be } \Rightarrow 540 - 540 = 0$$

$\Rightarrow$ The variances of the differences will be the sum of the variances of sample size $n_1 = 32$, and $n_2 = 50$;

$$\Rightarrow \sigma_{\bar{X}} \sigma_d = \sigma \sqrt{1/n_1 + 1/n_2} = 50 \sqrt{1/32 + 1/50} = 11.319$$

$$z_1 = \frac{-20}{11.319} = -1.77$$

$$z_2 = \frac{20}{11.319} = 1.77$$

$$\Rightarrow P(|\bar{X}_1 - \bar{X}_2| > 20) = 2 P(Z < -1.77) = 2(0.0384) = 0.0768$$

b) $z_1 = 5/11.319 = 0.44$

$$z_2 = 10/11.319 = 0.88$$

$$\Rightarrow P(-10 < \bar{X}_1 - \bar{X}_2 < -5) +$$

$$P(5 < \bar{X}_1 - \bar{X}_2 < 10) = 2(P(5 < |\bar{X}_1 - \bar{X}_2| < 10))$$

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$$= 2P(0.44 < Z < 0.88) = 2[P(Z < 0.88) - P(Z < 0.44)]$$

$$= 2(0.8106 - 0.6700) = 0.2812$$

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$n_1 = 25$	$n_2 = 36$	} taken from a normal population;
$\mu = 80$	$\mu = 75$	
$\sigma = 5$	$\sigma = 3$	

Sample mean computed from $n = 25$: $\bar{X}_1$

Sample mean computed from $n = 36$: $\bar{X}_2$

$$P(\bar{X}_1 - \bar{X}_2) \Rightarrow P(3.4 < \bar{X}_1 - \bar{X}_2 < 5.9)$$

$$\Rightarrow \bar{X}_1 \sim N(80, 5^2/25) = \bar{X}_1 \sim N(80, 1)$$

$$\Rightarrow \bar{X}_2 \sim N(75, 3^2/36) = \bar{X}_2 \sim N(75, 0.25)$$

$$\Rightarrow \bar{X}_1 - \bar{X}_2 \sim N(80 - 75, 1 + 0.25) = \bar{X}_1 - \bar{X}_2 \sim N(5, 1.25)$$

$\therefore$ The sampling distribution of the difference in sample mean $(\bar{X}_1 - \bar{X}_2)$;

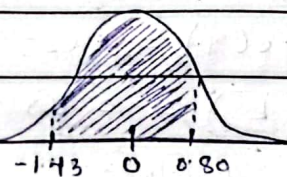
$$\Rightarrow \mu_{\bar{X}_1 - \bar{X}_2} = 5 \quad \bar{X}_1 - \bar{X}_2 \sim N(5, 1.25)$$

$$\Rightarrow \sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{1.25} = 1.12$$

$\therefore$ Convert to z-values;

$$\Rightarrow Z_1 = \frac{3.4 - 5}{1.12} = -1.43$$

$$\Rightarrow Z_2 = \frac{5.9 - 5}{1.12} = 0.80$$



$$\therefore P(-1.43 < Z < 0.80) = 0.7881 - 0.0764$$

$$= 0.7117 \times 100$$

$$= 71.17 \% \text{ chances}$$

Both of the samples are taken from two different populations, and are independent of each other.

From t-distribution table, for 14 degrees of freedom and a two-tailed probability of 0.05 (i.e. 0.025 in each tail), the critical value is :
 $t_{0.05, 14} = 2.145$

For a one-tailed probability of 0.10 with 10 degrees of freedom, the critical value is : $t_{0.10, 10} = 1.372$
 $-t_{0.10, 10} = -1.372$

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(9) For a cumulative probability of 0.995 with 7 degrees of freedom, the critical value is:

$$t_{0.995, 7} = \underline{\underline{3.499}}$$

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Q12.

a) From the t-distribution table, for 7 degrees of freedom:

$$P(T < 2.365) = 0.975$$

b) From the t-distribution table, for 24 degrees of freedom:

$$P(T > 1.318) = 0.10$$

c) From the t-distribution table, for 12 degrees of freedom:

$$P(T < 2.179) = 0.975$$

$$P(T < -1.356) = 0.10 \quad (\text{since } P(T > 1.356) = 0.10)$$

$$\therefore P(-1.356 < T < 2.179) = 0.975 - 0.10 = 0.875$$

d) Since the t-distribution is symmetric:

$$P(T > -2.567) = 1 - P(T < -2.567) = 1 - 0.01 = 0.99$$

Q13. Given a sample size of 24, the degrees of freedom $\nu = 23$;

a) From the t-distribution table, for 23 degrees of freedom:

$$P(T < -2.069) = 0.025$$

$$\therefore P(T < K) = 0.975 + 0.025 = 0.99$$

$\therefore$ From the table:

$$\Rightarrow K = t_{0.99, 23} = \underline{\underline{2.5}}$$

b) From the t-distribution table, for 23 degrees of freedom:

$$P(T < 2.807) = 0.995$$

$$\therefore P(T < K) = 0.995 - 0.095 = 0.9$$

$\therefore$ From the table:

$$\Rightarrow K = t_{0.9, 23} = \underline{\underline{1.319}}$$

c) This represents a two-tailed probability, so each tail has 0.05 probability.

$\therefore$ From the t-distribution table, for 23 degrees of freedom:

$$\Rightarrow K = t_{0.95, 23} = \underline{\underline{1.714}}$$

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Q14. Sample mean = 24

$$\text{Mean} = 20$$

$$\text{Sample size} = 9$$

$$\text{sample standard deviation} = 4.1$$

} from a normal population

We can perform a one-sample t-test since the population variance is unknown and the sample size is small ($n < 30$).

$$\Rightarrow \mu = 20$$

$$\bar{x} = 24$$

$$s = 4.1$$

$$n = 9$$

$$df = n - 1 = 8$$

$$\Rightarrow t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{24 - 20}{4.1/\sqrt{9}} = \frac{4}{4.1/3} = \frac{4}{1.3667}$$

$$t = 2.927$$

$\Rightarrow$ From the t-distribution table for $df = 8$,

At 5% significance level ($\alpha = 0.05$), the critical t-value is approximately ± 2.306 .

$$\Rightarrow \text{Our calculated } t = 2.927 > 2.306$$

$\therefore$ So, this sample mean is in the rejection region.

Q15. To test the claim that the mean saturated fat content is 0.5g, we perform a **one-sample t-test** since :

⇒ The population variance is UNKNOWN.

⇒ Sample size is small ($n=8$; $n < 30$)

⇒ The data is assumed to be from a normal distribution.

■ Null hypothesis (H_0) = $\mu = 0.5$ (claim is correct)

Alternative hypothesis (H_1) = $\mu \neq 0.5$ (mean is diff.)

$$\Rightarrow \bar{x} = \frac{0.6 + 0.7 + 0.7 + 0.3 + 0.4 + 0.5 + 0.4 + 0.2}{8}$$

$$\bar{x} = 3.8/8 = 0.475$$

$$\Rightarrow s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{(0.6-0.475)^2 + (0.7-0.475)^2 + (0.7-0.475)^2 + (0.3-0.475)^2 + (0.4-0.475)^2 + (0.5-0.475)^2 + (0.4-0.475)^2 + (0.2-0.475)^2}{7}$$

$$\Rightarrow s^2 = \frac{0.234}{7} = 0.0334 \quad ; \quad s = \sqrt{0.0334} = 0.183$$

$$\Rightarrow t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{0.475 - 0.5}{0.183/\sqrt{8}} = -0.386$$

⇒ Degrees of freedom, $df = n-1 = 8-1 = 7$

For a two-tailed test at $\alpha = 0.05$, the critical t-value $\approx \pm 2.365$.

∴ The calculated t-value (t-value = -0.386) falls within the interval (-2.365 to 2.365) ;

⇒ We **agree with claim** that the average saturated fat content is 0.5 grams.

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Q17. Minimum sample size n required;

$$\Rightarrow E = \text{margin of error} = 2$$

$$\Rightarrow \text{Population standard deviation} = \sigma = 5.8$$

$$\Rightarrow \text{Confidence level } 99\% ;$$

$$\Rightarrow Z_{\alpha/2} = 2.576$$

$$\Rightarrow n = \left(\frac{Z_{\alpha/2} / 2\sigma}{E} \right)^2 = \left(\frac{2.576 (5.8)}{2} \right)^2$$

$$\Rightarrow n = \left(\frac{14.9408}{2} \right)^2 = (7.4704)^2 \approx 55.83$$

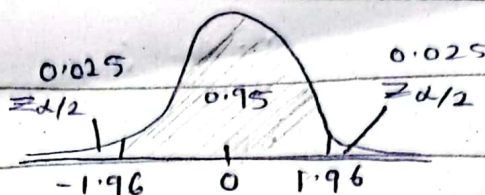
Since we can't take a fraction of a mouse, round up to the nearest whole number;

$$\therefore \underline{\underline{n = 56}}$$

Q18. A random sample of $n = 75$;

Since $n = 75$, $n > 30$, and the distribution is normal;

We need to construct a confidence interval of 95%;



$$\Rightarrow \text{Margin of Error} : E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 1.96 \left(\frac{0.0015}{\sqrt{75}} \right)$$

$$\Rightarrow E = 3.39 \times 10^{-4} = 0.000339$$

$\therefore$ For the confidence interval for the sample mean;

$$\Rightarrow \bar{X} - E < \mu < \bar{X} + E$$

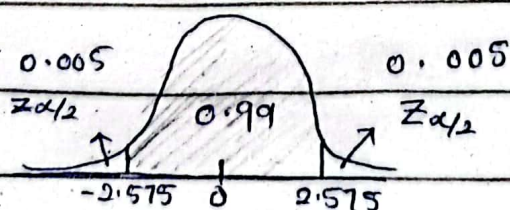
$$\Rightarrow 0.310 - 0.00039 < \mu < 0.310 + 0.00039$$

$$\Rightarrow 0.30961 < \mu < 0.31039$$

Q19. A random sample of $n = 100$;

Since $n = 100$, $n > 30$, and the distribution of measurements are approximately normal;

We need to construct a 99% confidence interval;



$$\Rightarrow Z_{\alpha/2} = 2.575$$

$$\Rightarrow E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2.575 \frac{(3900)}{\sqrt{100}} = 1004.25 = \text{margin error}$$

$\therefore$ For the confidence interval;

$$\Rightarrow \bar{x} - E < \mu < \bar{x} + E$$

$$\Rightarrow 23500 - 1004.25 < \mu < 23500 + 1004.25$$

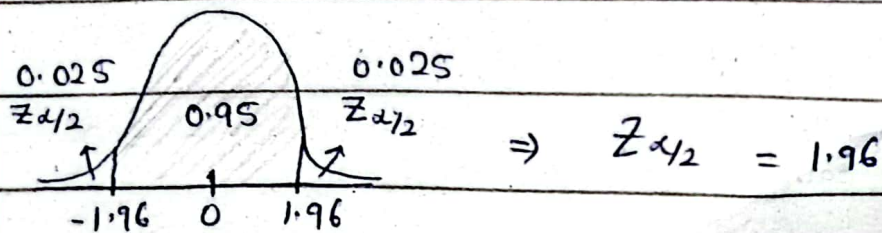
$$\Rightarrow 22495.75 < \mu < 24504.25$$

b) The maximum error (also called the margin of error) is the quantity we added/subtracted from the sample mean to create the interval.

$\Rightarrow$ We can assert with 99 % confidence that the possible error in the average number of kilometers driven by car owners in Virginia is no more than approximately 1005 km.

Q20. Sample of size $n = ?$

A confidence interval of 95 % is to be constructed;



$$E = \text{margin of error} = 15 = \frac{Z_{\alpha/2} \sigma}{\sqrt{n}}$$

$$\Rightarrow 15(\sqrt{n}) = 1.96(40)$$

$$\Rightarrow \sqrt{n} = \frac{1.96(40)}{15}$$

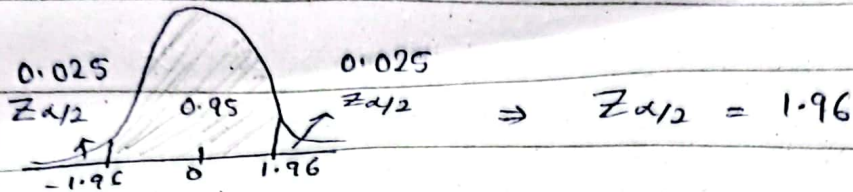
$$\Rightarrow n = \left(\frac{1.96(40)}{15} \right)^2 = 27.318$$

$\therefore$ A random sample of $n = 28$ will be needed.

Q21. A random sample of $n = 12$;

Since $n < 30$, and the distribution is normal
for the number of words typed per minute ;

We need to construct a 95 % confidence interval ;



$$\Rightarrow E = \text{margin of error} = \frac{z_{\alpha/2} \sigma}{\sqrt{n}} = \frac{1.96 (7.8)}{\sqrt{12}}$$

$$\Rightarrow E = 4.41$$

$\therefore$ For the confidence interval ;

$$\Rightarrow \bar{x} - E < \mu < \bar{x} + E$$

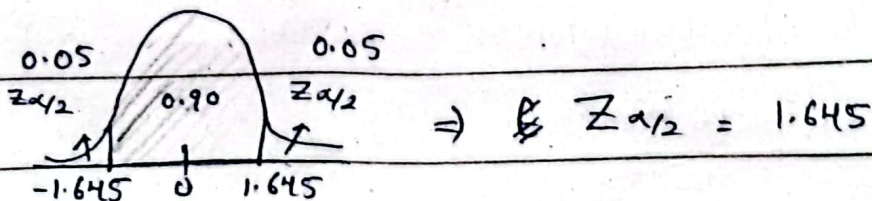
$$\Rightarrow 79.3 - 4.41 < \mu < 79.3 + 4.41$$

$$\Rightarrow 74.89 < \mu < 83.71$$

Q22. A random sample of $n = 12$;

Since $n < 30$, and the measurements are assumed
to be normally distributed ;

We need to construct a 90 % confidence interval ;



$$\Rightarrow E = \text{margin of error} = \frac{z_{\alpha/2} \sigma}{\sqrt{n}} = \frac{1.645 (1.5)}{\sqrt{12}}$$

$$\Rightarrow E = 0.71$$

∴ For the confidence interval :

$$\Rightarrow \bar{x} - E < \mu < \bar{x} + E$$

$$\Rightarrow 48.50 - 0.71 < \mu < 48.50 + 0.71$$

$$\Rightarrow \underline{47.79 < \mu < 49.21}$$

Q23. A random sample of $n = 15$;

Since $n < 30$, and the measurements represent a random sample from a normal population ;

$$\Rightarrow \bar{x} = \frac{3.4 + 2.5 + 4.8 + 2.9 + 3.6 + 2.8 + 3.3 + 5.6 + 3.7 + 2.8 + 4.4 + 4.0 + 5.2 + 3.0 + 4.8}{15}$$

$$\Rightarrow \bar{x} = 3.787 = \text{sample mean}$$

~~Formula of the sample standard deviation :~~ Sample standard deviation ;

$$\Rightarrow S = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

$$\Rightarrow S = 0.92$$

⇒ Using the prediction interval formula :

$$\Rightarrow \bar{x} \pm t_{\alpha/2} \cdot S \sqrt{1 + \frac{1}{n}}$$

⇒ For 95% prediction interval and $df = 14$;

$$t_{0.025, 14} = 2.145$$

$$\Rightarrow \sqrt{1 + \frac{1}{15}} = \sqrt{1.0667} = 1.033$$

$$\Rightarrow \text{Margin of error} = (2.145)(0.92)(1.033) = 2.04$$

$$\Rightarrow 3.787 - 2.04 < \mu < 3.787 + 2.04$$

$$\Rightarrow \underline{1.75 < \mu < 5.83}$$